MILITARY SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, FLEXIBLE, COAXIAL, 50 OHMS, M17/93-RG178 AND M17/93-00001

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cables described herein shall consist of this specification and the latest issue of MIL-C-17.

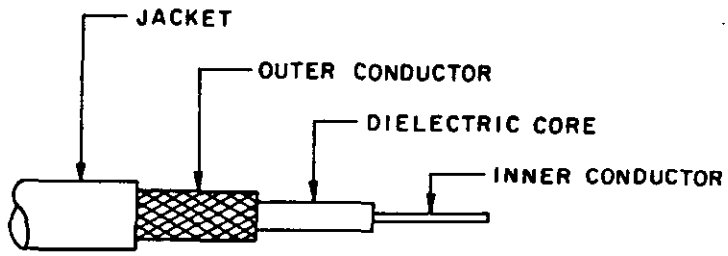


FIGURE 1. Configuration.

TABLE I. Description.

Components	Construction details
Inner conductor	Seven strands of silver-coated, annealed-copper-covered, steel wire, each strand, .004 inch diameter. Overall diameter: .012 inch ±.001.
Dielectric core	Type F-I: Solid, extruded PTFE. Diameter: .033 inch ±.002.
Outer conductor	Single braid of AWG size 38 silver-coated copper wire. Diameter: .054 inch, maximum. Coverage: 95.9%, nominal Carriers: 16 Ends: 3 Picks/inch: 25.0 ±10%
Jacket	Type IX: FEP for M17/93-RG178. Type XIII: PFA for M17/93-00001. Diameter: .071 inch ±.004.

Ⓖ denotes changes.

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ENGINEERING INFORMATION:

Continuous working voltage: 750 V rms, maximum.

Operating frequency: 3 GHz, maximum.

Velocity of propagation: 69.5 percent, nominal.

Power rating: See figure 2.

Operating temperature range:

-55°C to +200°C (M17/93-RG178).
-55°C to +230°C (M17/93-00001).

Inner conductor properties:

DC resistance (maximum at 20°C): 24.45 ohms per 100 feet.

Elongation: 10 percent, minimum.

Tensile strength: 50 $klb_f/inch^2$, minimum.

Engineering notes: This cable is useful in general purpose, high temperature applications (see connector series "SMA", "SMB", and "SMC" per MIL-C-39012).

REQUIREMENTS:

Dimensions, configuration, and descriptions: See figure 1 and table I.

Environmental and mechanical:

Visual and mechanical examination:

Eccentricity: 10 percent, maximum.

Adhesion of conductors:

Ⓖ Inner conductor to core: 1.0 pound, minimum; 4 pounds, maximum.

Ⓖ Cold bend: -55°C ±2°C.

Stress-crack resistance: +230°C ±5°C (for M17/93-RG178) or +250°C ±5°C (for M17/93-00001) for 96 hours; mandrel size 7 1/2 times the jacket diameter.

Dimensional stability: +200°C ±5°C.

Ⓖ Inner conductor from core: .125 inch, maximum.

Inner conductor from jacket: .312 inch, maximum.

Flammability: Applicable.

Weight: .625 pound per 100 feet, maximum.

Electrical:

Test frequency: 50 MHz to 3 GHz.

Spark test: 2,000 V rms, ±25 percent, -0 percent.

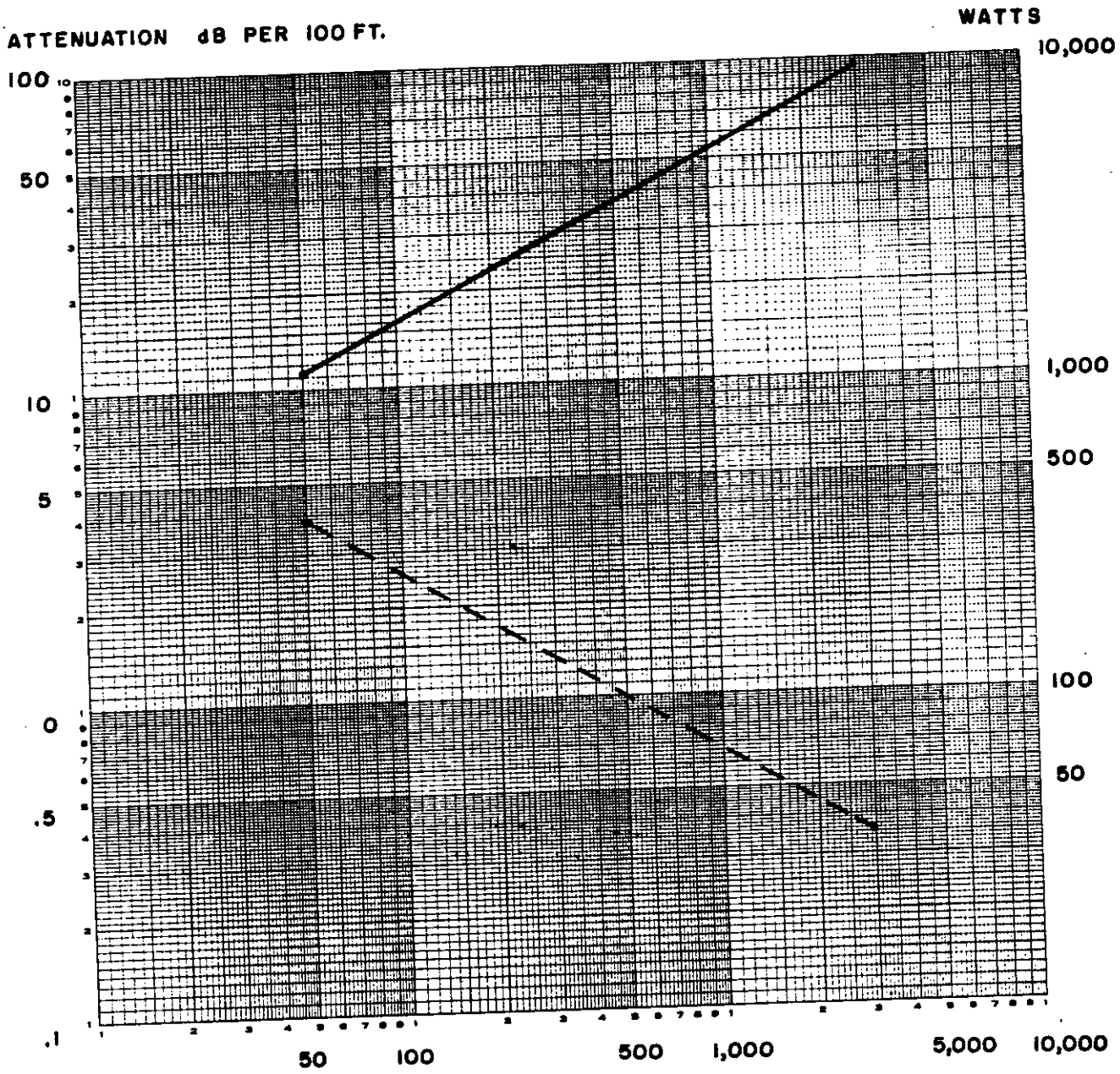
Voltage withstanding: 2,000 V rms, minimum.

Corona extinction voltage: 1,000 V rms, minimum.

Characteristic impedance: 50 ±2 ohms.

Attenuation: See figure 2.

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Attenuation	
MHz	dB
50	11.6
100	16
400	33
1000	52
3000	94

FIGURE 2. Power and attenuation.

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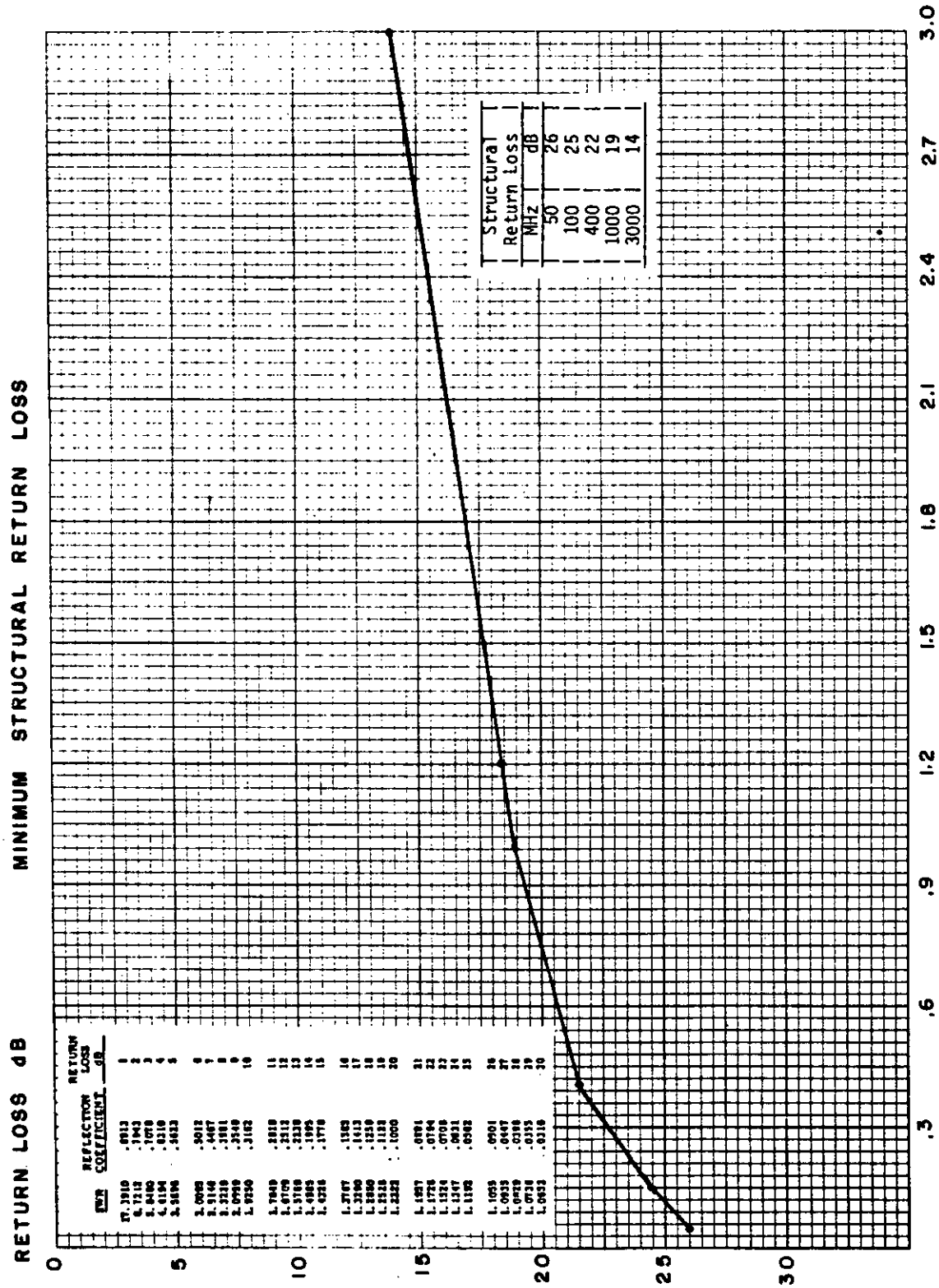


FIGURE 3. Structural return loss.

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Structural return loss: See figure 3.

Capacitance: 32 pF per foot, maximum.

Capacitance unbalance: Not applicable.

Transmission unbalance: Not applicable.

Mechanically induced noise voltage: Not applicable.

Time delay: Not applicable.

Contamination: Not applicable.

Part number: M17/93-RG178 (NATO preferred type NWR-34).
M17/93-00001.

Supersession data: See table II.

Ⓖ TABLE II. Cross reference of part number.

Part number	Superseded part number or type designation
M17/93-00001	M17/140-00001 per MIL-C-17/140 (canceled)

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH
Air Force - 11, 17, 99
DLA - ES, IS

User activities:
Army - AR, AT, ME
Navy - AS, MC, OS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - CR

(Project 6145-0875-1)

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